Q15 resolved (Tier 1, item 28)

TPC-H Q15 was dropped from the entire project — all 21-query statistics exclude it, and every document has had to explain the gap. The stated reason was that it "requires a view/temp-table construct that needed special handling in the collection script".

That is accurate but it was never a hard problem. The spec form wraps the query in `CREATE VIEW revenue0 ...` / `DROP VIEW revenue0`, which a single-statement `EXPLAIN (ANALYZE, BUFFERS, FORMAT JSON)` harness cannot express. A CTE is a single statement and semantically identical.

The subtlety that makes it correct rather than merely convenient

A naive `WITH revenue0 AS (...)` is **not** equivalent to the view. The CTE is referenced twice (once in the join, once in the `max()` subquery). PostgreSQL 12+ materialises a CTE referenced more than once, computing it once and sharing the result. A view is inlined at every reference, so the spec form evaluates the aggregate twice. Different plan, different cost, different runtime.

`AS NOT MATERIALIZED` restores view semantics. Measured on the local SF1 instance:

form	node_count	plan_depth	root_cost	rows_processed
spec form (real `CREATE VIEW`)	13	7	**306862.09**	641,812
CTE `NOT MATERIALIZED` (ships as `q15.sql`)	13	7	**306862.09**	611,857
CTE `MATERIALIZED`	11	6	153957.94	335,922

The `NOT MATERIALIZED` rewrite reproduces the view form's plan **exactly** — same node count, same depth, same total cost to the cent. The materialised variant is a visibly different plan and would have been the wrong substitution. The 4.7% gap in `rows_processed` between the two identical-plan forms is run-to-run variation in how many parallel workers the executor was granted, not a semantic difference.

Both the view form and the correct CTE cost roughly **2x** the materialised form, so had Q15 been collected with a plain `WITH`, its runtime would have been systematically understated.

Status

`scripts/collection/queries/q15.sql` now exists and runs under the standard harness. The workload is **22 queries**, not 21.

What this does *not* do: there is no Q15 row in `tpch_dataset_corrected_v2.csv`, because that would require re-running the five EC2 instances, which are terminated. Q15 is available for all *new* measurement — the elasticity probe, plan-structure features, and the arm64 laptop baseline — and must be collected for every machine in the Tier 2 expansion. Until then, cross-machine statistics remain 21-query and should say so for that reason rather than the old one.